Luke Phan

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EDUCATION

University of Illinois Urbana-Champaign *Expected May 2027*

Bachelor of Science in Aerospace Engineering GPA: 3.54/4.00

Relevant Coursework: Applied CFD (Compressible RANS turbulence modeling), Compressible Flow, Incompressible Flow (XFLR5), Numerical Methods (Python), Aerospace Control Systems (Python), Thermodynamics

WORK EXPERIENCE

Combined Metals Company – *Engineering Intern (Python)* Elgin, IL | May 2025 – Aug. 2025

- Developed Python-based ETL pipelines and analysis tools to process aerospace alloy data, reducing engineering analysis time by 25%
- Applied statistical analysis and machine learning (Random Forest, K-means, PCA) to analyze high-dimensional metallurgical datasets, identifying dominant process–property relationships and variability trends
- Analyzed large-scale metallurgical datasets to identify trends in yield, strength, and process variability for aerospace OEMs (GE, Rolls-Royce)
- Presented data-driven findings to engineering leadership, preventing $40K+ in rejected material

PROJECTS

Applied CFD Final Project – Airfoil Ground Effect Study (SU2, Gmsh, ParaView, Python) Jan. 2026 – May 2026

- Built full 2D S1223 airfoil ground-effect CFD workflow: geometry, meshing, solver setup, and post-processing
- Ran SU2 viscous sims at $M = 0.3$, $Re = 3 \times 10^6$, and $\alpha = 1°$ to evaluate aerodynamic coefficient trends
- Conducted mesh refinement study (3.4K to 136K cells), assessed convergence, runtime, and coefficient behavior
- Automated data processing and visualization in Python for aerodynamic coefficients, mesh-quality metrics, and report-ready figures

Formula SAE – Illini Electric Motorsports – Aero Team Jan. 2025 – Jan. 2026

- Participated in aerodynamic design discussions, interpreting CFD results including lift/drag coefficients and pressure coefficient (Cp) distributions
- Gained exposure to CFD workflows in STAR-CCM+, including geometry setup, solver selection (k-ω, k-ε), and convergence monitoring through residuals
- Observed simulation-driven design (multi-element bullhorn configurations) and impact on aero performance
- Assisted in manufacturing structural components using machining, bonding processes, and hand tools

Personal Projects – CFD and FEA (ANSYS Fluent & Mechanical) Jan. 2025 – May 2025

- Simulated compressible and incompressible flows (airfoil, nozzle, mixing tee) using ANSYS Fluent, analyzing pressure distribution, lift/drag, and flow structures
- Performed mesh refinement and ensured solution convergence through residual monitoring and result consistency
- Compared CFD results with analytical solutions and experimental correlations to validate simulation accuracy
- Post-processed simulation data using Python to extract aerodynamic coefficients and visualize flow fields

SAB P-51 Project (Siemens NX) – Wing Team Sep. 2024 – May 2025

- Designed wing structure parts using sheet metal CAD in NX, incorporating manufacturing standards
- CNC-machined sheet metal parts and assembled a scaled-down model P-51D with a 20+ member team

SKILLS

Engineering & Simulation: SU2, ANSYS Fluent, ANSYS Mechanical, Gmsh, ParaView, STAR-CCM+

Programming: Python (NumPy, Pandas, Plotly, scikit-learn), Jupyter

CAD: Creo, Siemens NX, Fusion 360

Productivity: Microsoft Excel, SharePoint, Word, PowerPoint, Teams, Outlook